Calibrate Cross-Validation Grid Size

Description

Compares candidate spatial-grid sizes using occupied-cell density and fold balance, then marks the finest eligible candidate.

Usage

```
compute_cross_validation_grid_calibration(
  data_locations = NULL,
  candidate_grid_cell_sizes_km = NULL,
  n_folds = 5L,
  n_repeats = 1L,
  occupancy_criterion = "median",
  target_locations_per_cell = NULL,
  lower_quantile_probability = 0.25,
  max_fold_location_difference = 1L,
  max_fold_sample_difference = Inf,
  seed = 900723L
)
```

Arguments

`data_locations`	Location table returned by [build_cross_validation_location_table()].
`candidate_grid_cell_sizes_km`	Numeric vector of unique positive candidate grid-cell widths in kilometres.
`n_folds`	Single integer greater than one and no greater than the location count.
`n_repeats`	Single positive integer used to evaluate deterministic origin shifts.
`occupancy_criterion`	Character scalar: '"minimum"', '"lower_quantile"', or '"median"'.

`target_locations_per_cell`	Positive numeric occupancy target. Defaults to 'n_folds' when 'NULL'.
`lower_quantile_probability`	Probability used by '"lower_quantile"', between zero and '0.5'. Defaults to '0.25'.
`max_fold_location_difference`	Maximum allowed difference between the largest and smallest fold location counts within any repeat. Defaults to '1L'.
`max_fold_sample_difference`	Maximum allowed difference between fold sample counts within any repeat. Defaults to 'Inf' so it is diagnostic until calibrated explicitly.
`seed`	Single non-negative integer assignment seed. Defaults to '900723L'.

Details

Candidate assignments use [build_spatial_cross_validation_assignments()]. Eligibility requires the configured occupancy statistic and both balance thresholds. Among eligible candidates, the smallest cell size is selected.

Value

Tibble with one row per candidate grid size, occupancy and balance diagnostics, eligibility, selection status, and a single 'selected' flag when at least one candidate is eligible.

Examples

```
data_locations <-

  tibble::tibble(

    location_id = letters[1:6],

    coord_x_km = c(0, 1, 2, 10, 11, 12),

    coord_y_km = rep(0, 6),

    n_samples = rep(1L, 6),

    row_indices = as.list(seq_len(6L))

  )

compute_cross_validation_grid_calibration(

  data_locations = data_locations,

  candidate_grid_cell_sizes_km = c(1, 10),

  n_folds = 3L
```

)